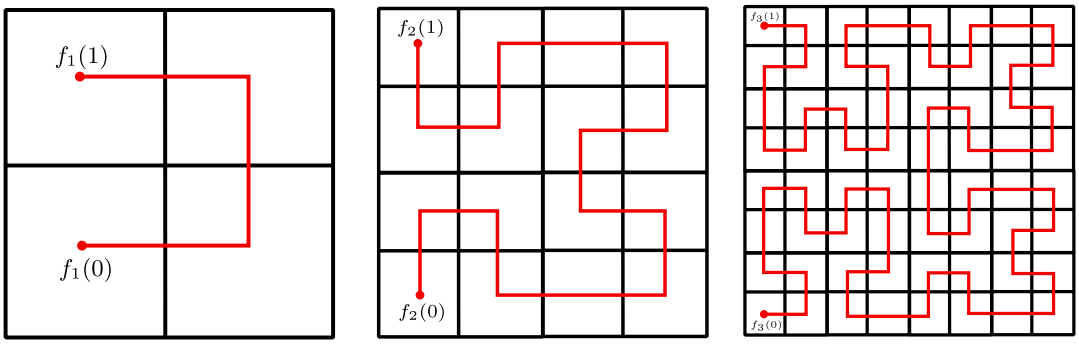
All solutions to graded problems must be typed in LaTeX. All other problems can be typed or written, but must be completed. Graded problems must be organized in numerical order. All problem sets (both graded and non-graded) must be uploaded to Canvas as a single PDF document. Students can find the grading scale in our Syllabus.

Graded

Exercise 1. Let $\mathscr{T}$ and $\mathscr{T}'$ be compact Hausdorff topologies on a set X. Prove that either $\mathscr{T} = \mathscr{T}'$ or the two topologies are not comparable.

Exercise 2. Let $f_1, f_2, f_3, \ldots$ be a sequence of functions from $[0,1]$ to $[0,1] \times [0,1]$ with the rule for each f_n suggested by the diagrams



Prove that the sequence $\{f_n\}$ converges uniformly to some function $f : [0,1] \to [0,1] \times [0,1]$ and that f is surjective.

Non-Graded

Exercise 3. Prove that if X is a set with a strict total order in which every closed interval is compact in the order topology, then X has the least upper bound property.

Exercise 4. Let $\mathbb{R}_K$ denote the K-topology on $\mathbb{R}$.

1. Show that the interval $[0,1]$ is not compact as a subspace of $\mathbb{R}_K$.
2. Show that $\mathbb{R}_K$ is connected.
3. Show that $\mathbb{R}_K$ is not path connected.